

Technical Requirement Document

Here is the Technical Requirement Document (TRD) based on the provided Functional Requirement Document (FRD):

TR-DLT-001: Load Customer and Order Data into Delta Tables

Related Functional Requirement(s): FR-DLT-001

Objective: Implement a Delta Live Table to load customer and order data from CSV files into delta tables.

Target Cluster Configuration:

- Cluster Name: Databricks Cluster for DLT
- Databricks Runtime Version: 10.4.x-scala2.12
- Node Type: Standard_DS3_v2
- Driver Node: Standard_DS3_v2
- Worker Nodes: 2-4 Standard_DS3_v2
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: delta-lake, spark-csv

Source Data Details:

- Dataset Name: customerdata
- Location: /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata
- Format: CSV
- Description: Customer data in CSV format
- Dataset Name: orderdata
- Location: /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata
- Format: CSV
- Description: Order data in CSV format

Target Data Details:

- Output Name: customer
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.customer
- Format: Delta
- Description: Customer delta table
- Output Name: order

- Location: gen_ai_poc_databrickscoe.sdlc_wizard.order
- Format: Delta
- Description: Order delta table

Job Flow / Pipeline Stages:

- Read customer CSV data into a delta table named "customer"
- Read order CSV data into a delta table named "order"
- Add a new column "TotalAmount" to the "order" delta table

Data Transformations / Business Logic:

- Step: Read CSV data into delta tables
- Description: Read customer and order CSV data into delta tables
- Transformation Logic: Use spark.read.csv to read CSV data and write delta format
- Step: Add new column "TotalAmount" to "order" delta table
- Description: Calculate TotalAmount by multiplying PricePerUnit and Qty columns
- Transformation Logic: Use spark.sql to add new column with calculated value

Error Handling and Logging:

- Log errors to a log file
- Use try-except blocks to handle exceptions

TR-DLT-002: Cleanse Customer and Order Data

Related Functional Requirement(s): FR-DLT-002

Objective: Implement data cleansing logic to remove null and duplicate records from customer and order delta tables.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details: Same as TR-DLT-001

Target Data Details: Same as TR-DLT-001

Job Flow / Pipeline Stages:

- Remove rows with null values from "customer" and "order" delta tables
- Remove duplicate rows from "customer" and "order" delta tables

Data Transformations / Business Logic:

- Step: Remove null values
- Description: Remove rows with null values from delta tables
- Transformation Logic: Use spark.sql to filter out null values
- Step: Remove duplicates
- Description: Remove duplicate rows from delta tables

- Transformation Logic: Use spark.sql to remove duplicates

Error Handling and Logging: Same as TR-DLT-001

TR-DLT-003: Create ordersummary Table and Load Data

Related Functional Requirement(s): FR-DLT-003

Objective: Implement a Delta Live Table to create an "ordersummary" table and load data into it by joining customer and order delta tables.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: customer
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.customer
- Format: Delta
- Description: Customer delta table
- Dataset Name: order
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.order
- Format: Delta
- Description: Order delta table

Target Data Details:

- Output Name: ordersummary
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.ordersummary
- Format: Delta
- Description: Ordersummary delta table

Job Flow / Pipeline Stages:

- Create "ordersummary" table if it does not exist
- Join "customer" and "order" delta tables on "CustId" column
- Load joined data into "ordersummary" table

Data Transformations / Business Logic:

- Step: Join customer and order delta tables
- Description: Join customer and order delta tables on CustId column
- Transformation Logic: Use spark.sql to join tables
- Step: Load data into ordersummary table
- Description: Load joined data into ordersummary table
- Transformation Logic: Use spark.sql to load data into SCD type 2 table

Error Handling and Logging: Same as TR-DLT-001

TR-DLT-004: Update ordersummary Table on Customer Data Change

Related Functional Requirement(s): FR-DLT-004

Objective: Implement logic to update "ordersummary" table when there is a change in the customer table.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details: Same as TR-DLT-003

Target Data Details: Same as TR-DLT-003

Job Flow / Pipeline Stages:

- Identify changes in "customer" table
- Update "ordersummary" table by making old records inactive and new records active
- Update "StartDate" and "EndDate" columns accordingly

Data Transformations / Business Logic:

- Step: Identify changes in customer table
- Description: Identify changes in customer table
- Transformation Logic: Use spark.sql to identify changes
- Step: Update ordersummary table
- Description: Update ordersummary table by making old records inactive and new records active
- Transformation Logic: Use spark.sql to update SCD type 2 table

Error Handling and Logging: Same as TR-DLT-001

TR-DLT-005: Create customeraggregatespend Table and Load Data

Related Functional Requirement(s): FR-DLT-005

Objective: Implement a Delta Live Table to create a "customeraggregatespend" table and load aggregated data from the "ordersummary" table.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: ordersummary
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.ordersummary
- Format: Delta
- Description: Ordersummary delta table

Target Data Details:

- Output Name: customeraggregatespend

- Location: gen_ai_poc_databrickscoe.sdlc_wizard.customeragggregatespend
- Format: Delta
- Description: Customeragggregatespend delta table

Job Flow / Pipeline Stages:

- Create "customeragggregatespend" table if it does not exist
- Aggregate "TotalAmount" column from "ordersummary" table by grouping by "Name" and "Date" columns
- Load aggregated data into "customeragggregatespend" table

Data Transformations / Business Logic:

- Step: Aggregate data
- Description: Aggregate TotalAmount column by grouping by Name and Date columns
- Transformation Logic: Use spark.sql to aggregate data
- Step: Load data into customeragggregatespend table
- Description: Load aggregated data into customeragggregatespend table
- Transformation Logic: Use spark.sql to load data into delta table

Error Handling and Logging: Same as TR-DLT-001

TR-DLT-006: Implement Delta Live Tables

Related Functional Requirement(s): FR-DLT-006

Objective: Implement Delta Live Tables to automate the data pipeline.

Target Cluster Configuration: Same as TR-DLT-001

Job Flow / Pipeline Stages:

- Implement Delta Live Tables to automate the data pipeline

Data Transformations / Business Logic: Same as previous TRs

Error Handling and Logging: Same as TR-DLT-001